A. Twice

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: implementation | https://codeforces.com/problemset/problem/2037/A

Kinich wakes up to the start of a new day. He turns on his phone, checks his mailbox, and finds a mysterious present. He decides to unbox the present.

Kinich unboxes an array a with n integers. Initially, Kinich's score is 0. He will perform the following operation any number of times:

- Select two indices i and j ($1 \leq i$ lt $j \leq n$) such that neither i nor j has been chosen in any previous operation and $a_i = a_j$. Then, add 1 to his score.

Output the maximum score Kinich can achieve after performing the aforementioned operation any number of times.

Input

The first line contains an integer t ($1 \leq t \leq 500$) — the number of test cases.

The first line of each test case contains an integer n ($1 \leq n \leq 20$) — the length of a.

The following line of each test case contains n space-separated integers $a_1, a_2, \ldots, a_n$ ($1 \leq a_i \leq n$).

Output

For each test case, output the maximum score achievable on a new line.

Example

Input
5 1 1 2 2 2 2 1 2 4 1 2 3 1 6 1 2 3 1 2 3

Output
0 1 0 1 3

Note

In the first and third testcases, Kinich cannot perform any operations.

In the second testcase, Kinich can perform one operation with $i=1$ and $j=2$.

In the fourth testcase, Kinich can perform one operation with $i=1$ and $j=4$.